

Olist Product Quality Analysis

Can Olist predict — and prevent — low-quality customer experiences?

Business Analytics Capstone · Olist Dataset · 2026

96,478

Orders analysed

4.14★

Platform avg score

13.2%

Low-score rate (1–2★)

AUC 0.733

Full model performance

What we cover today

Three questions. Two ML approaches. Five actionable policies.

BQ1

Category & vendor quality audit

Which categories carry the most complaint risk?

BQ2

Logistics vs product quality

Are bad reviews caused by shipping or defective goods?

BQ3

Listing completeness & satisfaction

Do better product pages lead to higher scores?

ML1

Supervised: predicting bad reviews

Logistic regression + ROC + odds ratios

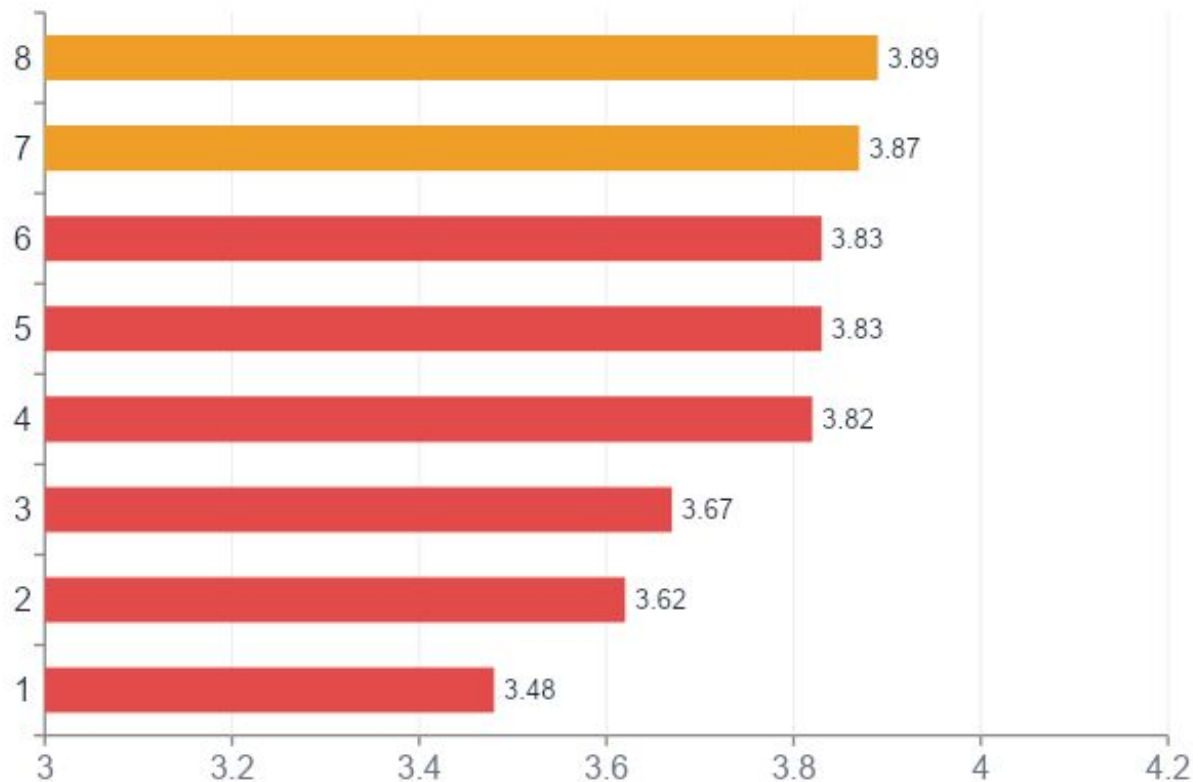
ML2

Unsupervised: seller segmentation

Manual business segmentation + hierarchical clustering

BQ1 · Category Quality Audit

Which product categories carry the highest complaint risk? Platform avg: 4.02★ | Min 50 reviews per category



High risk >20% low-score

3 categories



Watch list 15–20%

6+ categories



Healthy <15%

48 categories

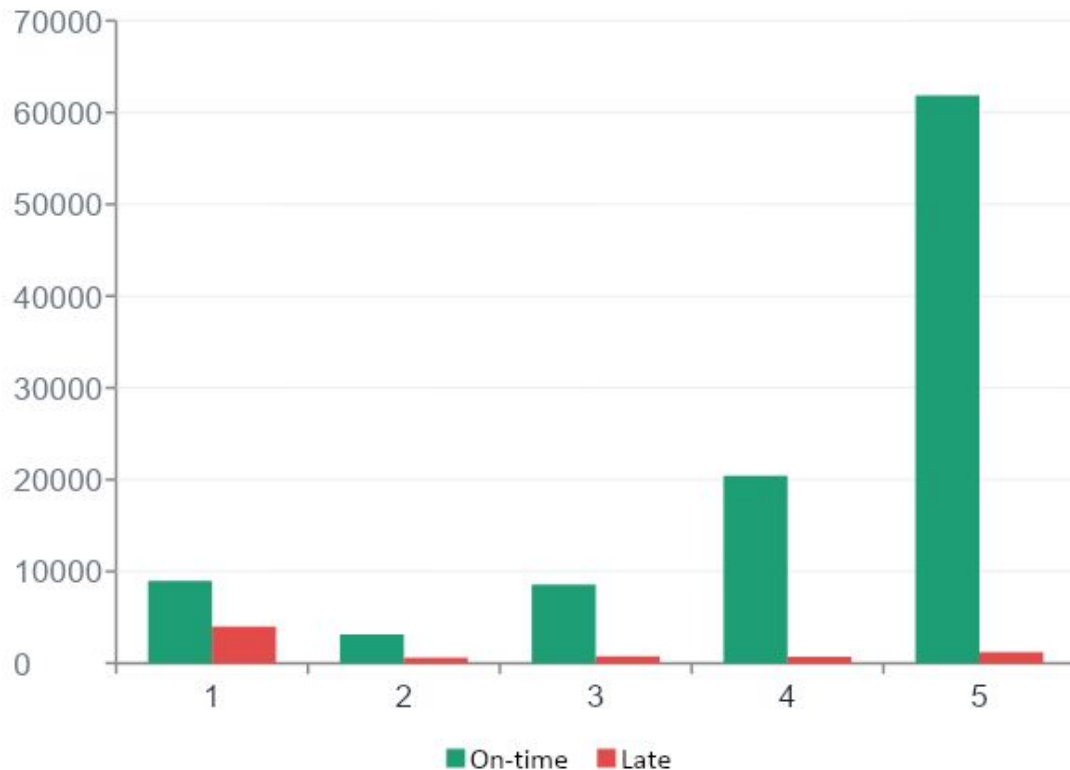
Policy Recommendation

Categories >20% low-score rate:
mandatory packaging guidelines + quality tests before new seller onboarding.

15–20%: watch list with quarterly review targets.

BQ2 · Logistics vs Product Quality

Late deliveries nearly halve the average review score — and drive 63% of all 1–2 star reviews



On-time avg score

4.28★

9.5% low-score · 60.1% five-star
88,652 orders

Late avg score

2.54★

54.6% low-score · 16.3% five-star
7,826 orders

Logistics penalty

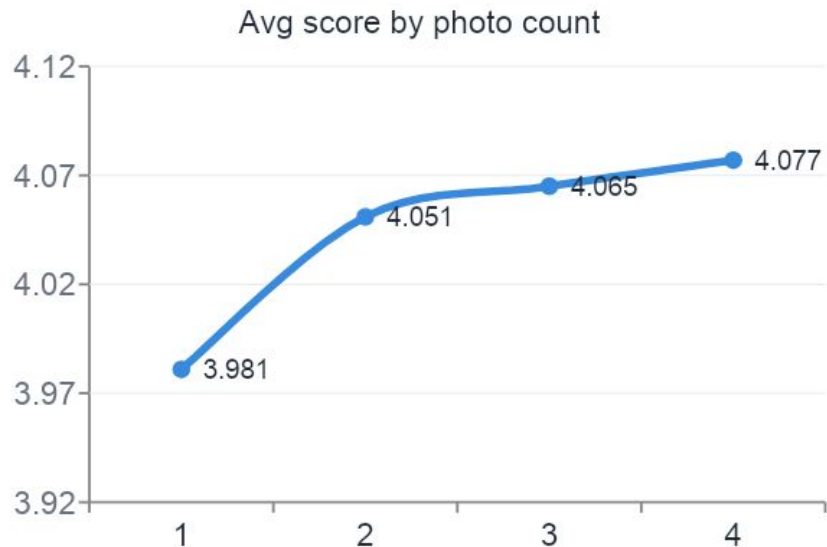
-1.74★

score drop from late delivery

Policy: dynamic EDD buffer + separate logistics scorecard from product scorecard

BQ3 · Listing Completeness & Satisfaction

Better listings lift scores — but the gains plateau quickly after 2 photos and 300 characters



+0.070★

1 photo → 2–3 photos

+0.051★

<300 → 300+ char desc

Gate

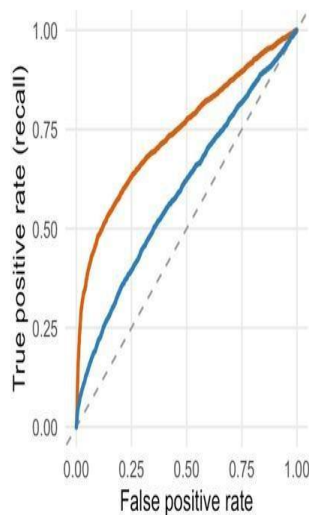
Min 2 photos + 300 chars to publish

ML1 - Supervised Learning: Can We Predict Bad Reviews?

Logistic regression on 80/20 train-test split · Target: review score of 1 or 2 stars

Predicting Low-Quality Customer Experiences

ROC curves on the unseen 20% test set



Model — Full customer experience — Product/listing risk

Target: review score of 1 or 2 stars

The orange (full model) curve sits higher and further from the diagonal than the blue (product-only) curve. A perfect model would reach the top-left corner.

Q: Can Olist predict bad reviews?

A: Yes. Full model AUC = 0.733 — useful predictive ability.

Q: Does product info alone predict poorly?

A: Yes. Product/listing model AUC = only 0.591 — barely above chance.

Q: What does AUC mean?

A: Higher = better. A perfect model scores 1.0; random guessing scores 0.5.

Why logistic regression? Binary target, interpretable odds ratios, transparent stakeholder decisions.

ML1 - Operational Scorecard: What Happens in Practice?

The model flags 18.2% of orders, captures 51.7% of actual low reviews, and is 2.85× more accurate than random selection

Metric	Full customer-experience model	Product/listing model only
Orders flagged	3,459 (18.2%)	7,045 (37.0%)
Low-review orders caught	1,291 (51.7%)	1,248 (50.0%)
Low-review rate in flagged group	37.3%	17.7%
Risk lift vs random	2.85×	1.35×

Focused workload

Flag just 18.2% of orders for intervention — not 100%. Customer service effort is concentrated where risk is highest.

Majority captured

51.7% of actual 1–2 star reviews are in that 18.2% flagged group — more than half of all complaints identified proactively.

2.85× risk lift

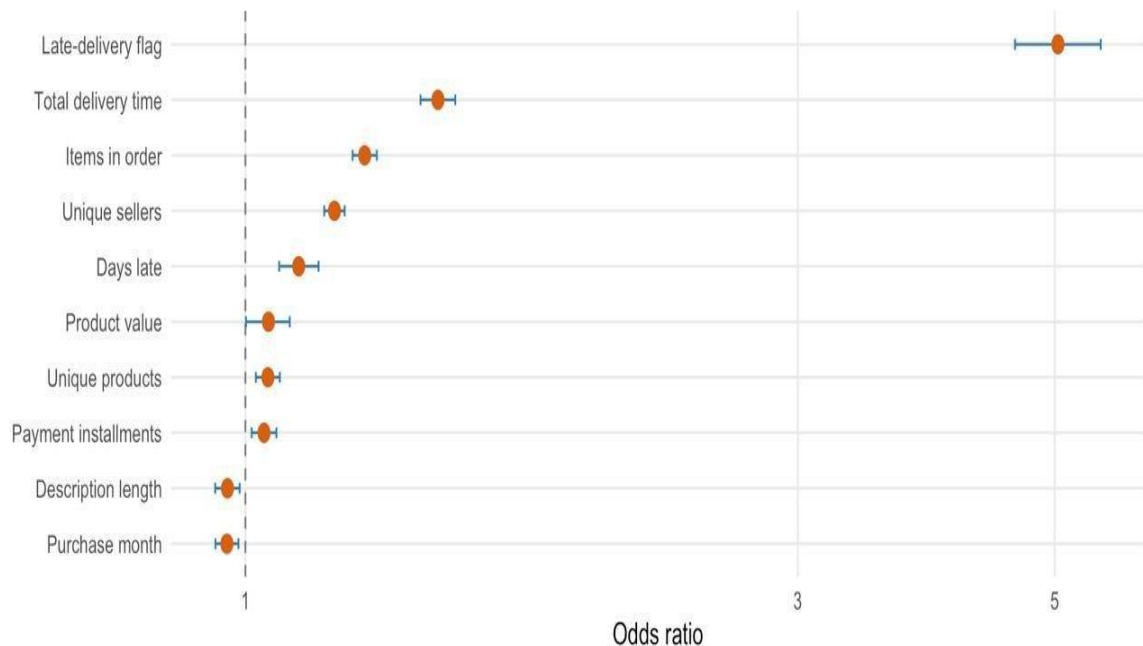
A flagged order is 2.85× more likely to receive a low review than a randomly selected order. The model is meaningfully precise.

ML1 - What Drives Dissatisfaction? Odds Ratios

Late delivery is the single strongest signal — associated with ~5× higher odds of a 1–2 star review

Strongest Signals of a Low Review

Odds ratios from the full customer-experience model



Numeric features: one-standard-deviation increase. Late flag: late versus on time.

Late delivery flag ~5×

Largest risk factor. A late order has ~5× higher odds of a low review.

Delivery time ~2×

Longer total delivery time increases risk, independent of the late flag.

Items in order ~1.7×

Complex multi-item orders generate more friction and higher complaint risk.

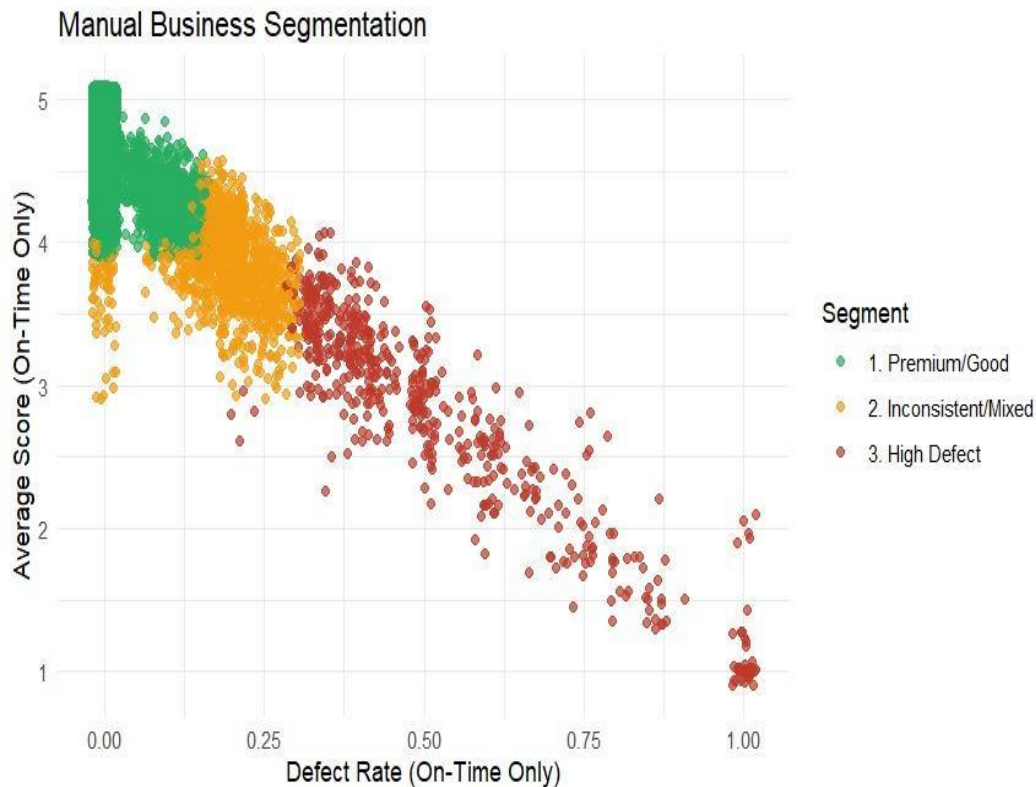
Description length <1×

Longer descriptions reduce risk — better-informed customers have fewer complaints.

These are associations, not proven causal relationships.

ML2 · Unsupervised: Manual Business Segmentation

Three interpretable segments defined by on-time defect rate and average score — directly actionable for vendor management



1. Premium / Good

4.55★ · 64.4% of products · 2.6% defect rate

→ Protect & incentivise — platform backbone

2. Inconsistent / Mixed

3.85★ · 23.9% of products · 19.1% defect rate

→ Monitor & improve — needs targeted support

3. High Defect

2.82★ · 11.7% of products · 51.4% defect rate

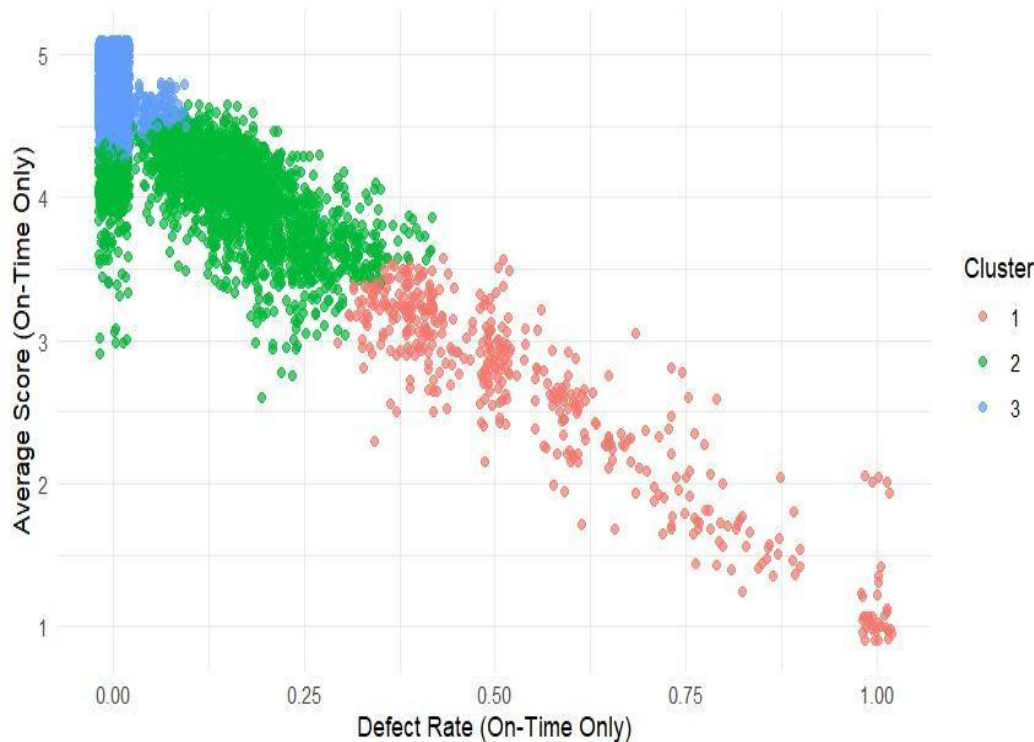
→ Review for removal or probationary SLA

Segmentation based on on-time defect rate and avg review score (on-time deliveries only, to isolate product quality from logistics).

ML2 - Algorithmic Validation: Ward's Method

The algorithm independently confirms the three-segment structure — validating the manual business segmentation

Algorithmic Clustering (Hierarchical / Ward's)



What the algorithm found



Same 3-cluster structure

Ward's method independently arrives at 3 clusters that mirror the manual Premium/Inconsistent/High-Defect segments.



Silhouette peak at k=2

Mathematically, k=2 is the most stable partition. k=3 is chosen for business interpretability — the third segment is the actionable 'high defect' group.



84.6% variance retained (PCA)

Two principal components capture 84.6% of total dataset variance, confirming the 2D scatter accurately represents the full feature space.



Overlap is expected

Continuous e-commerce metrics produce overlapping cluster boundaries — the distinct centers and tails are what matter for segmentation.

Conclusions & Recommendations

Five findings. Five policies. All grounded in data.

BQ1

6 categories exceed 20% low-score threshold

→ Quality SLA: mandatory tests before new seller onboarding

BQ2

Late delivery causes -1.96 ★ penalty (63% of 1★ reviews)

→ Dynamic EDD buffer + separate logistics from product scorecard

BQ3

2+ photos & 300+ char descriptions lift scores

→ Portal gate: minimum listing standards before publishing

ML1

Full model AUC 0.743 · flags 18.2% · catches 51.7%

→ Deploy as real-time risk flag for proactive customer recovery

ML2

11.7% of products are high-defect (51.4% defect rate)

→ Differentiated vendor strategy: protect stars, mentor middle, review risky

Supervised learning allows Olist to move from reacting to complaints toward proactively protecting customer satisfaction.